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NPTEL ONLINE CERTIFICATION COURSE

SUSTAINABLE TRANSPORTATION SYSTEMS

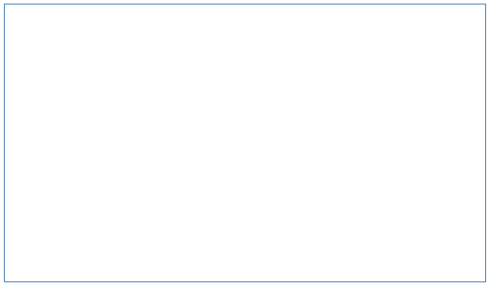
Lecture 22: Introduction of land use

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Contents

- Land use
- Connectionn between land use & transportation
- Integration of transport system into land use planning
- Concepts related to land use
- Land use strategies for sustainable transport



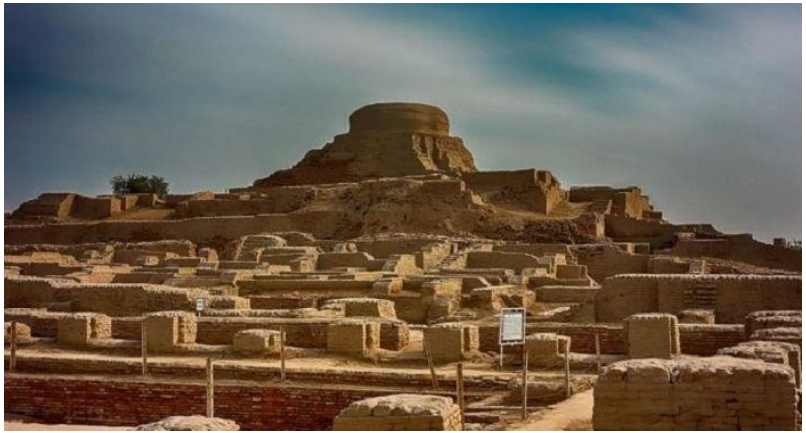
Historical perspective of land use: Indus valley civilization

Do you think land use planning is a modern time concept?

No, even in ancient civilization, land use planning was done.

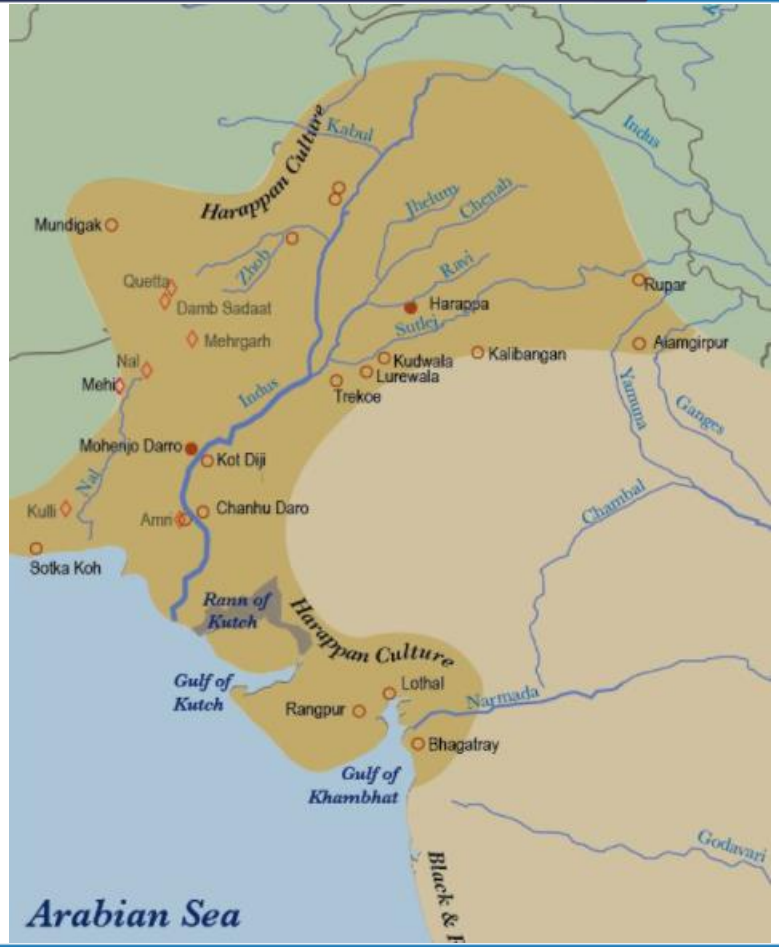
Residential areas, gathering areas and storage facilities were found in Indus valley civilization.

Roads were crossed at 90-degree angles.



Source: learnculturalindia.net

Historical perspective of land use & transportation: Indus valley civilization



- Most of sites were found either at riverbank or near sea.
- Rivers and sea used to provide easy navigation channels for trade activities.

Source: pininterest.com

Historical perspective of land use & transportation: Grant Trunk road

- Route between Kolkata to Kabul was developed during or before Mauryan empire.
- Later many kings renovated & renamed it. Finally, GT road named by Britishers during colonial time.
- This road connected many important cities.
- Many Sarai (rest houses) and public amenities were developed along the road.
- Later many new cities developed on this route.



Source: timesofindia.com

Concept of land use

- Land use is the term used to describe the human use of land.
- Represents the economic and cultural activities that are practiced at a given place.
- Public and private lands frequently represent very different uses.

Land use & transportation 1/2

- Land use & transportation are closely interlinked.
- Transportation systems are developed as per land use and vice-versa.

  - In Delhi metro, Red line of metro line was developed firstly due to high population density in nearby area.
  - When city sprawls further, new transportation systems such as feeder bus services or extension of existing metro is done to accommodate the changes.

Land use & transportation 2/2



- 10-25% of land is occupied by road transport infrastructure in urban areas.
- At the same time, a lot less space for pedestrian and NMT (non-motorized transport) infrastructure provided.

source: GIZ, wsp.com

Land use & transport: Connection

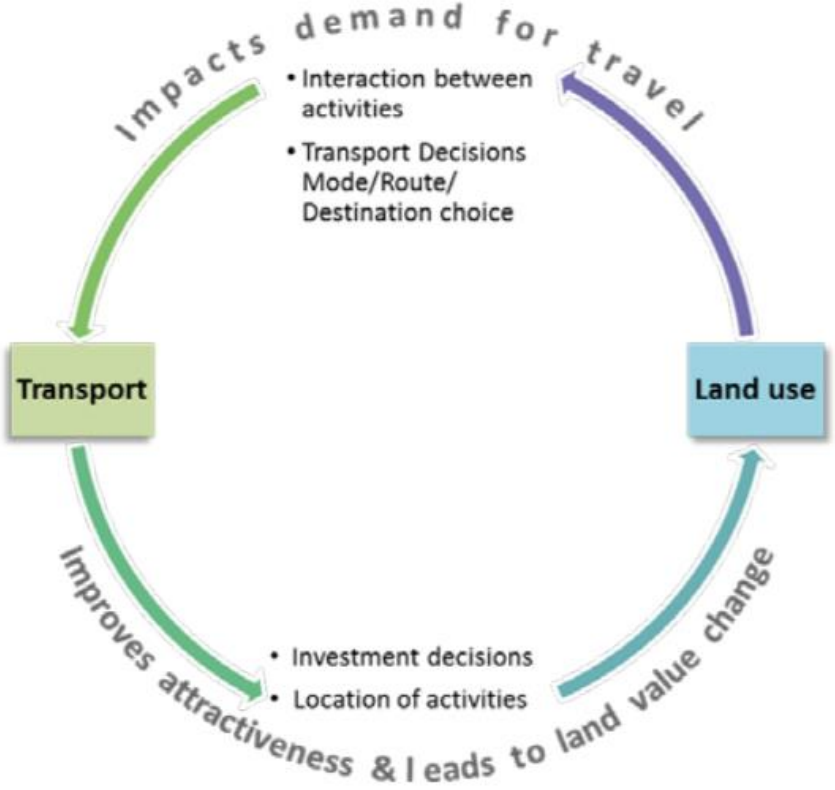


Image: Interconnection between land use & transport

Source: UNDP

Mechanism for land use & transport integration

Integrating Land use and transport in the planning process

- Policy and plan integration - vertical
- Policy and plan integration - horizontal

Institutional Integration and legal Mechanism

- Lead agency
- Plan area and horizon

Financing

- Funding plan preperation
- Funding the plan

Capacity Building

- Awareness generation
- setting up of transport section in the development authorities

Stakeholder Involvement

- Identification of stakeholders
- Setting up process for consultation

Source: UNDP

Advantage of land use & transport integration

- Sustainable use of urban land (low intensity uses in ecologically and culturally important locations; high intensity of use in locations that can support it)
- Reduced vehicle emissions and higher quality of living environment
- Less time spent in travel and therefore higher productivity as well as quality of life
- Greater access to public transport and thus to jobs

Impact of transportation system on land use

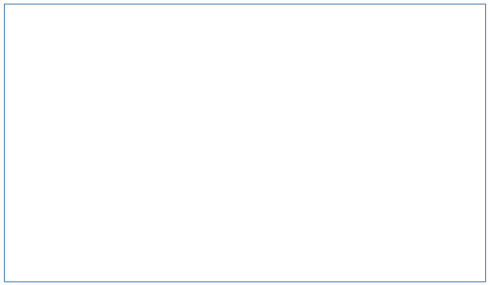
As transportation system passes through any locality, land/property values increase rapidly, and commercial activities grows in near by areas.

Ex.- Ahmedabad-Mumbai proposed bullet train passes through many small town/cities, it is expected that industrial and commercial activities will increase in these region which will further help in employment generation

Impact of land-use change on environment

- Land use changes occur constantly and at many scales

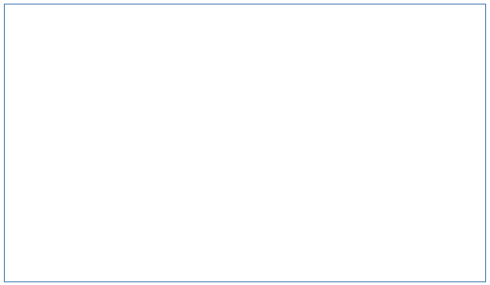
- It can have specific and cumulative effects on
  - Air and water quality
  - Watershed function
  - Generation of waste
  - Extent and quality of wildlife habitat
  - Climate
  - Human health



Example of impacts after land use change

If an agricultural land is changed into road, what would the possible impacts on local environment?

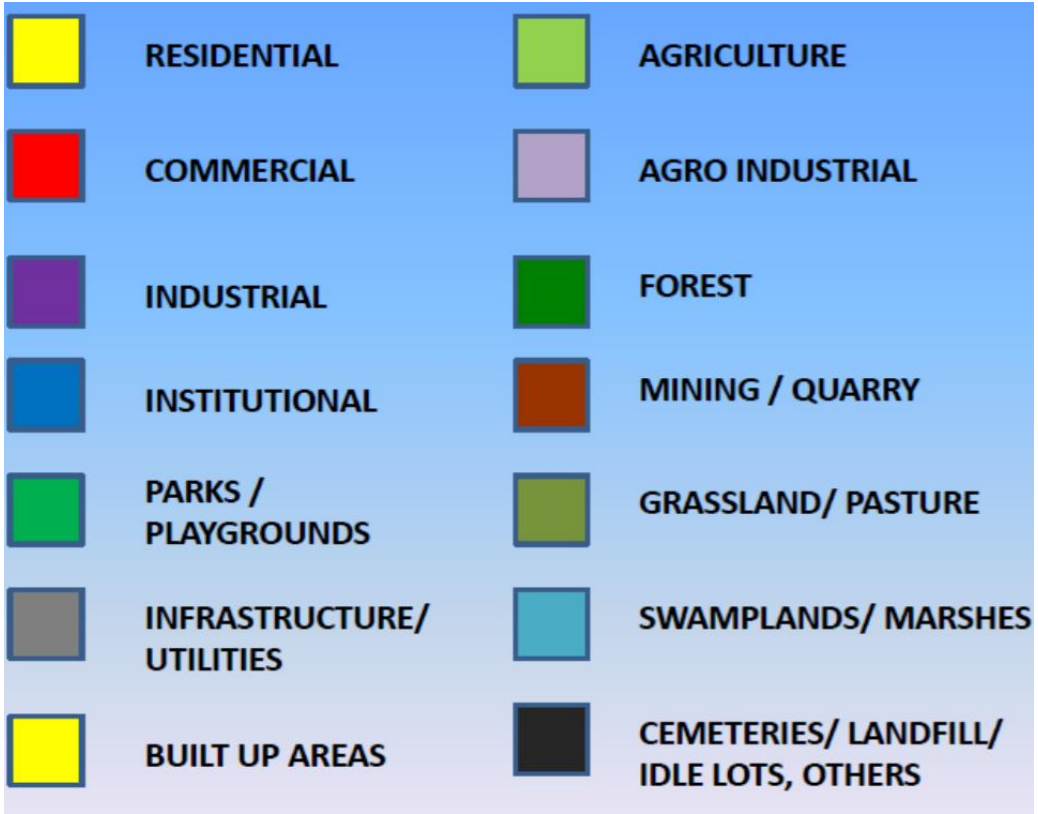
- Decrease in air quality
- Noise pollution
- Impacts on local vegetation
- In case of high pollution, adverse impacts on human health
- Livelihood of local population



Land use planning

- Rational and judicious approach of allocating available land resources to different land using activities and for different functions consistent with the overall development vision/goal of a particular city.

General color coding for land use map



In land use, transportation systems are considered under infrastructure/utilities category.

Land use map

- Land use map represents the various land uses within the study area

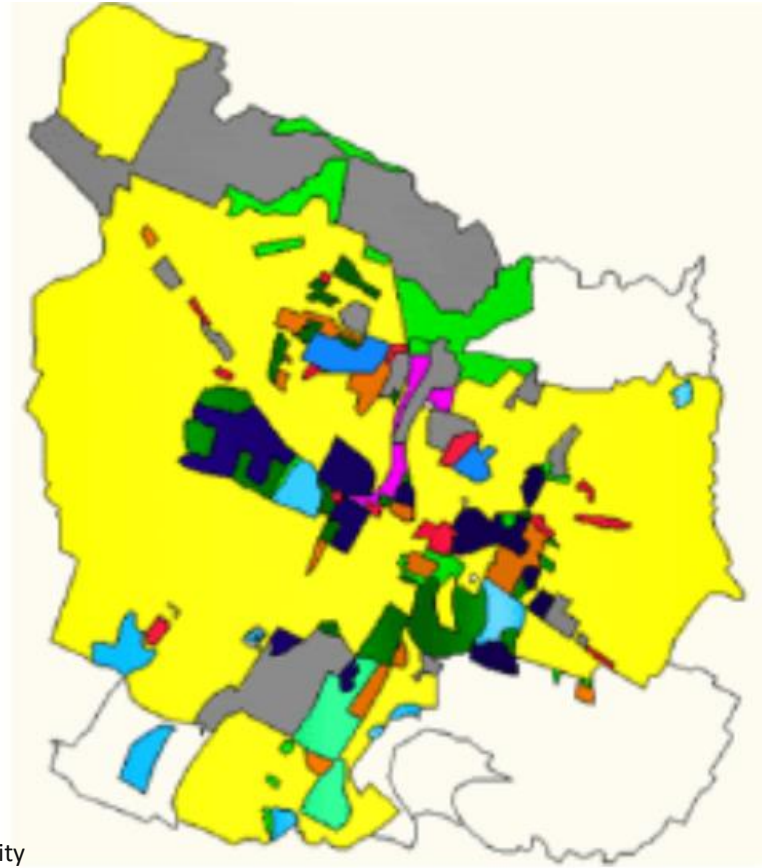
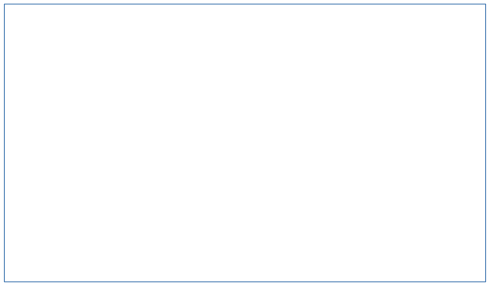


Image: Land use map of Mysore city, Karnataka

Source: Source: Mysore Urban Development Authority

Land use planning & transport: traditional approach

- Traditional focus was given to road design
- More infrastructure for cars
- More space for motorized vehicles
- Unsustainable approach



Land use & transport: Sustainable approach 1/2



Image source: GIZ

Land use & transport: Sustainable approach 2/2

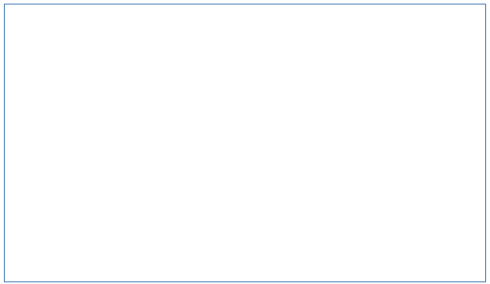
- Land use can be better with public and non-motorized transport.
- Land use should be done in a way that it promote NMT & public transport rather then giving priority to car and personal vehicles.



Image source: cyclepromotionfund.org

Land use planning strategies to reduce adverse impact of growth of transport sector

- Avoid sprawl
- Encouraging mixed Land Use
- Dense, compact development
- Transit oriented development (TOD)
- Good urban design
- Not too much land for roads and parking



Impact on environment of promoting NMT/PT

	Share (%) of public transport, walking and cycling	CO_2 emissions (kg per capita per year)
Houston	5%	5690 kg
Montreal	26%	1930 kg
Madrid	49%	1050 kg
London	50%	1050 kg
Paris	54%	950 kg
Berlin	61%	774 kg
Tokyo	68%	818 kg
Hongkong	89%	378 kg

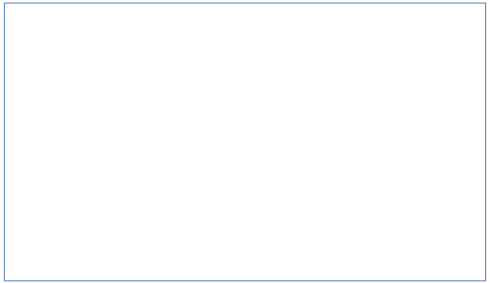
NMT: Non-Motorized Transport

PT-Public Transport

Source: UITP

Concepts related to land use

- Reversible-Irreversible land use
- Multiple land use
- Compatible-non compatible land use
- Best land use



Reversible land use

- Inherent features and characteristics of the land have not been considerably altered or modified
- Soil horizon, landform, and structure remain intact
- Land can be reverted to its former use or original condition
- Opposite of it, will be termed as irreversible land use

Ex: Reversible land use

- Temporary agriculture by many tribal societies are example of reversible land use.
- Forest land is converted into agricultural land for time being.
- After few seasons, land is gain abandoned and forest cover regrows.

Irreversible land use in transportation

- Most of the transport infrastructure has a very long lifetime
- Changes made during this time, will create permanent changes in the land use on site and around the site.
- Generally economic activities grows rapidly along the transportation corridor or terminal.

Multiple land use

- Combining different land uses, whether reversible or irreversible, in an orderly and desirable pattern
- Land can indeed have more than one use and uses can be combined in different ways

Example of multiple land use in transportation

- Many transportation hubs are also a commercial center.
- It is generally seen that many guest houses and hotels are constructed to attract travels from near by transport hub.



Image showing, MG road metro station in Gurugram. Multiple malls and shopping complex is developed around the metro station.

Image source: hinsdustantimes.com

Compatible land use

- Land use is considered compatible if it is compatible with land type as well as other land uses in case of multiple land use.

Do you think developing an industrial setup in fertile agriculture land is compatible or developing an airport in an eco-sensitive location is compatible land use?

No, it isn't

Best land use

- The use of land which generates the maximum profit without negative impacts (or with minimum consequences) especially on the environment
- Land should be used in such a manner consistent with its natural qualities to maximize its productivity and adhere to the principles of sustainable development.
- Utilizing land in a manner that is beneficial to both man and environment.

Industries with need of high-water demand near water sources or steel plants near coal or iron ore mine is best use of land?

Yes, it is.

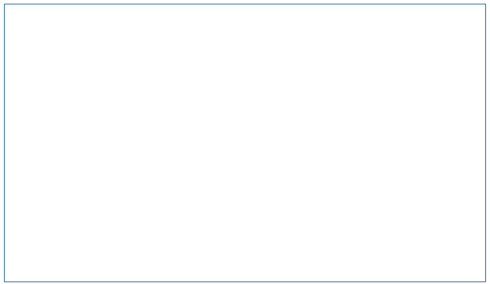
Example of poor land use: Amravati

- Amravati is new capital city of Andhra Pradesh after bifurcation of state.
- It is situated near Krishna river.
- Land is very fertile agricultural land with high crop yield.
- Environmentalists argue that development of whole new city will have high impacts on environment and example of poor land use.



Conclusion

- Land use should be done in a manner that optimum use of land is ensured.
- Land use and transportation impacts each other.
- A suitable land use planning may lead to better transport system.



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Thank You